25901547 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCSTA202 - PROBABILITY DISTRIBUTIONS AND STATISTICAL INFERENCE

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **7** questions

Each question carries **2** marks

1. Mention one application of the Type II Beta distribution. [K] / [CO1]
2. What happens to the Negative Binomial distribution when $r = 1$r = 1, where r is the number of successes. [U] / [CO1]
3. A lot consists of 20 bulbs, out of which 6 are defective. If 6 bulbs are chosen at random, find the probability that exactly 2 bulb is defective. [U] / [CO1]
4. State the variance of a Type I Beta random variable. [K] / [CO1]
5. Write one limitation of Bernoulli's Law of Large Numbers. [U] / [CO2]
6. Can Chebychev's Inequality be applied without knowing the distribution? [U] / [CO2]
7. How are the ranks assigned in the Wilcoxon signed rank test in one sample case? [K] / [CO3]
8. Write the test statistic used in two sample test for variance. [U] / [CO3]
9. Comment on the importance of normal distribution in large sample test. [K] / [CO3]

10. Distinguish between parametric and non parametric test. [U] / [CO3]

[2x7 = 14]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **6** marks

11. Define the log normal distribution. Also write the relation between log-normal and normal distribution. [U] / [CO1]

12. Derive the mean and moment generating function of Gamma distribution. [U] / [CO1]

13. In a quality control process, the probability that a product passes inspection is 0.8. Find the probability that 3 failures occur before 4 successes. Also find the expected number of failures before 4 successes. [U] / [CO1]

14. An unbiased coin is tossed 100 times. Let X be the number of heads occurs with E(X) = 50 and V(X) = 25. Find P(30 < X < 70) using Tchebycheff's inequality. [A] / [CO2]

15. Explain sign test. State clearly its assumptions. [U] / [CO3]

16. A political analyst claims that 55% voters favour a policy.In a sample of 300 voters, 150 favour the policy. Test whether the analyst's claim is valid at 5% significance level. [A] / [CO3]

[6x4 = 24]

Part C

Essay Questions

Answer any **1** questions

Each question carries **12** marks

17. Show by using CLT that binomial and Poisson tend to normal distribution for large n. [A] / [CO2]

18. A test was conducted on two batches of students and their scores are given below: [A] / [CO3]

Batch 1	3	4	8	6	2	5
Batch 2	9	7	5	10	8	6

Use appropriate non parametric test to check whether there is significant difference between the test scores of the two batches at 5% level of significance. Table value is 5.

[12x1 = 12]